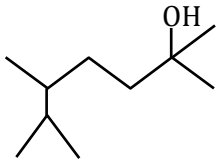
EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Introduction of Alcohols

1. Which is the correct IUPAC name for the following compound?



(A) 4–isopropyl–1, 1–dimethyl–1–pentanal
(B) 5–isopropyl–1, 1–dimethyl–2–hexanol
(C) 1,1,4,5–tetramethyl–1–hexanol
(D) 2,5,6–trimethyl–2–heptanol

CAE079

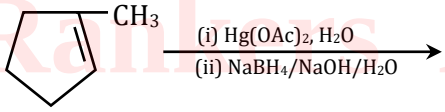
2. The IUPAC name of ethyl methyl ether is
(A) 1–methoxy ethane
(B) ethoxy methane
(C) 2–methoxy ethane
(D) 1–ethoxy methane

CAE080

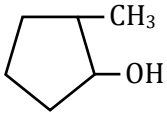
3. Which of the following is a tertiary alcohol
(A) $(CH_3)_2$ CH CH_2 OH
(B) $CH_3CH_2CH_2CH_2OH$
(C) $CH_3CH_2CH_2OH$
(D) $(CH_3)_3COH$

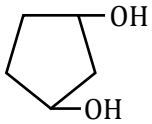
CAE081

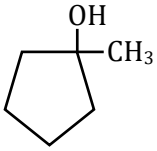
Method of preparation of Alcohol

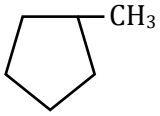
4. 

A, A is :

(A) 

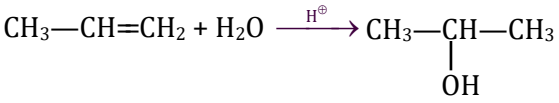
(B) 

(C) 

(D) 

CAE082

5. Select the incorrect option for the following reaction:



(A) This is an example of Nucleophilic addition reaction of alkene
(B) In the first step, protonation of alkene takes place to form carbocation
(C) In the second step, Nucleophilic attack of water takes place on carbocation
(D) In the last step deprotonation takes place to form an alcohol

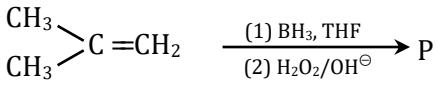
CAE083

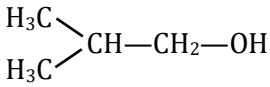
6. Which of the following statement is correct for hydroboration oxidation ?
(A) It follows Markovnikov's rule, and it is an anti-addition
(B) Effectively it follows Markovnikov's rule, and it is a syn-addition
(C) It does not follow Markovnikov's rule, and it is an anti-addition
(D) It does not follow Markovnikov's rule, and it is a syn-addition

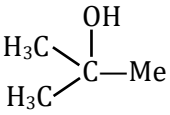
CAE084

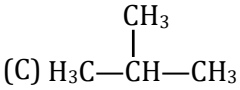
Rankers Academy JEE

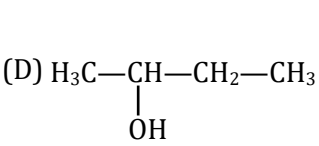
For More Join: @IITJEE_Advanced

7. 

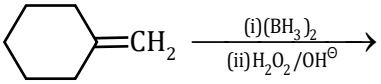
(A) 

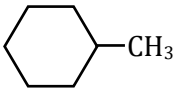
(B) 

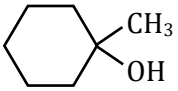
(C) 

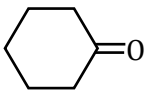
(D) 

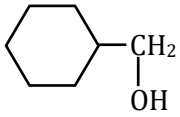
CAE085

8.  P; Product (P) in the reaction is :

(A) 

(B) 

(C) 

(D) 

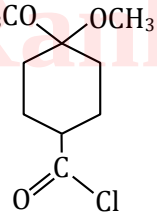
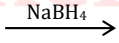
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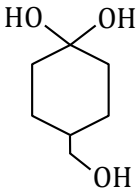
9. $H_2C=CH-COOH \xrightarrow{LiAlH_4}$ Product, is :

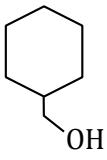
(A) $CH_3-CH_2-CH_2OH$
(B) CH_3-CH_2-CHO
(C) $CH_2=CH-CH_2OH$
(D) CH_3-CH_2-COOH

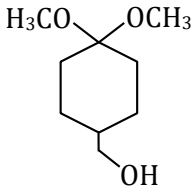
CAE087

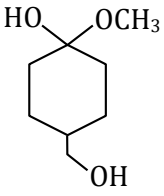
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10.   (X), Product (X) is :

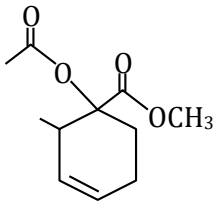
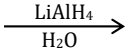
(A) 

(B) 

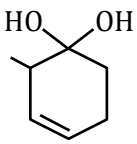
(C) 

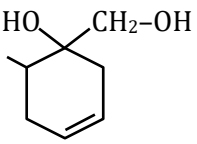
(D) 

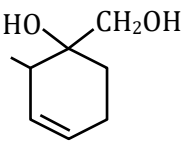
CAE088

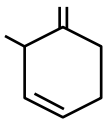
11.   (A)

Find out 'A' of the reaction:

(A) 

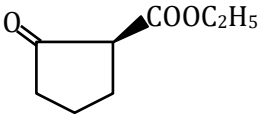
(B) 

(C) 

(D) 

CAE089

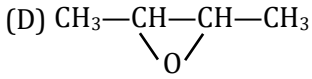
For More Join: @IITJEE_Advanced

12.  $\xrightarrow{LiAlH_4}$ $P_1 + P_2$; Products (P_1 and P_2) of the reaction are :

(A) Racemic (B) Diastereomers (C) Meso (D) Optically pure

CAE090

13. Which of the following compound gives tertiary alcohol with CH_3MgBr?

(A) C_6H_5CHO (B) $C_2H_5CO_2CH_3$ (C) C_2H_5COOH (D) 

CAE091

Reaction of Alcohol and ethers

14. $CH_3-\overset{O}{\overset{\|}{C}}-O-H + CH_3{}^{18}OH \xrightarrow{H^\oplus}$ Product,

Product can be :

(A) $CH_3-\overset{O}{\overset{\|}{C}}-O-CH_3$ (B) $CH_3-\overset{{}^{18}O}{\overset{\|}{C}}-O-CH_3$ (C) $CH_3-\overset{{}^{18}O}{\overset{\|}{C}}-\underset{18}{O}-CH_3$ (D) $CH_3-\overset{O}{\overset{\|}{C}}-\underset{18}{O}-CH_3$

CAE093

15. Arrange the following alcohols in the increasing order of esterification :

(a) sec-butyl alcohol (b) tert-Butyl alcohol (c) n-Butyl alcohol

(A) c > b > a (B) c > a > b (C) a > b > c (D) b > a > c

CAE094

16. The reaction

$CH_3-\overset{O}{\overset{\|}{C}}-OC_2H_5 + n\text{-}C_3H_7OH \xrightarrow{n\text{-}C_3H_7ONa} C_2H_5OH + CH_3COOC_3H_7(n)$

is known as :

(A) Esterification (B) Double decomposition

(C) Transesterification (D) None of these

CAE095

17. $CH_3-\overset{O}{\overset{\|}{C}}-OH + R_1OH \xrightarrow{H^\oplus} R_2-\overset{O}{\overset{\|}{C}}-O-Et$, R_1 & R_2 can be :

(A) –Me, –Et (B) –Et, –Me (C) –Me, –Me (D) –Et, –Et

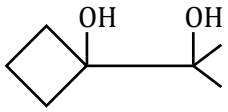
CAE096

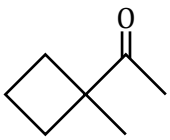
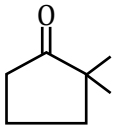
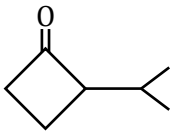
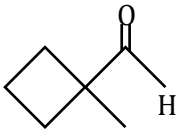
18. Esterification can be carried in the medium :

(A) Alkaline (B) Acidic (C) Neutral (D) All of the above

CAE097

19. Find out major product of following reaction :

 $\xrightarrow{H^\oplus}$ Major product

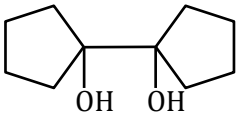
(A)  (B)  (C)  (D) 

CAE098

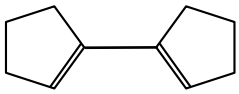
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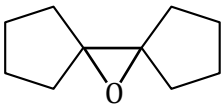
For More Join: @IITJEE_Advanced

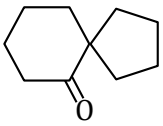
20. The structure of the product formed in the reaction given below is:

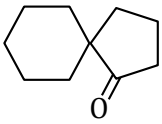


$\xrightarrow[\Delta]{H^{\oplus}}$ Product

(A) 

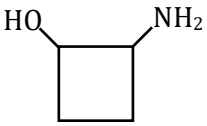
(B) 

(C) 

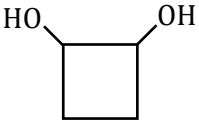
(D) 

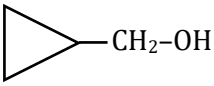
CAE099

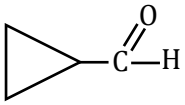
21. Major product of the reaction is :

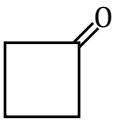


$\xrightarrow{HNO_2}$ Major product

(A) 

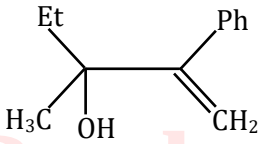
(B) 

(C) 

(D) 

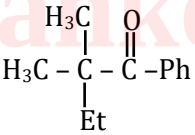
CAE100

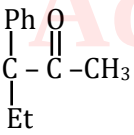
22. For the given reaction major product is :

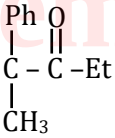


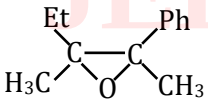
$\xrightarrow[\Delta]{H^{\oplus}}$ Major product

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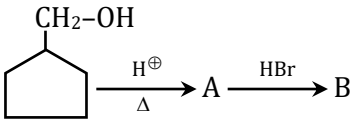
(A) 

(B) 

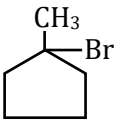
(C) 

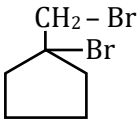
(D) 

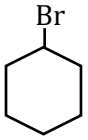
CAE101

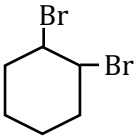
23. 

What is the structure of B?

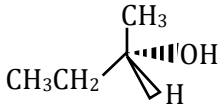
(A) 

(B) 

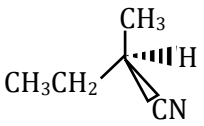
(C) 

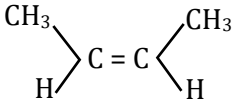
(D) 

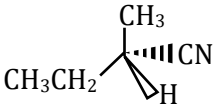
CAE103

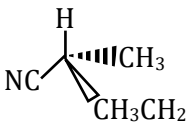
24. 

$\xrightarrow{PBr_3}$ $\xrightarrow[DMF]{KCN}$ Product , the product is :

(A) 

(B) 

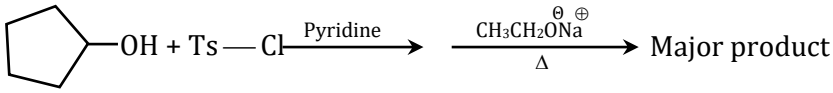
(C) 

(D) 

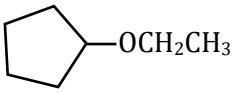
CAE104

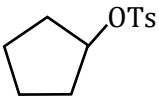
For More Join: @IITJEE_Advanced

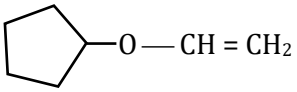
25. What would be the major product of following reactions ?



(A) 

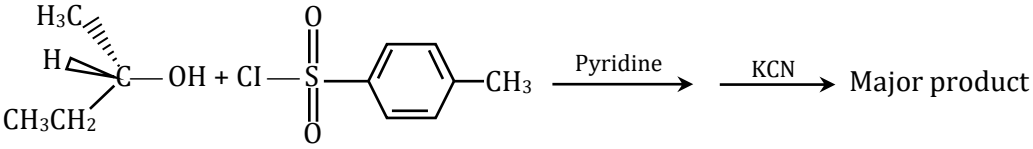
(B) 

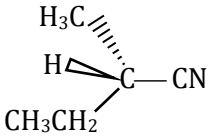
(C) 

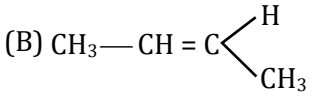
(D) 

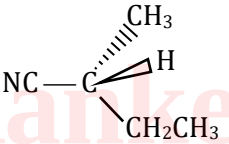
CAE105

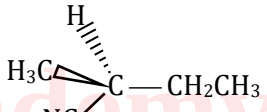
26. Find the major product of following reaction :



(A) 

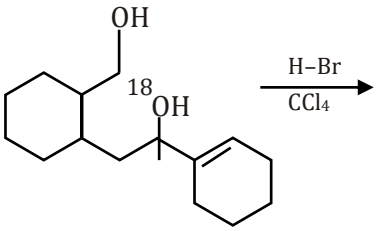
(B) 

(C) 

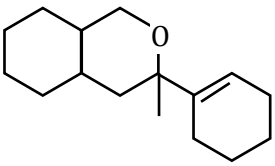
(D) 

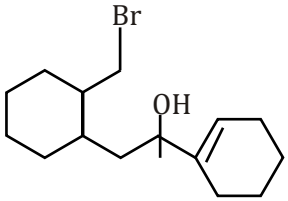
Rankers Academy JEE

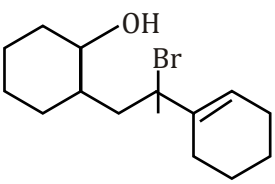
CAE106

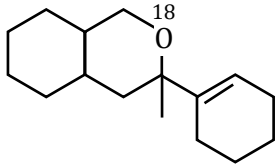
27. 

Major product obtained in the reaction is :

(A) 

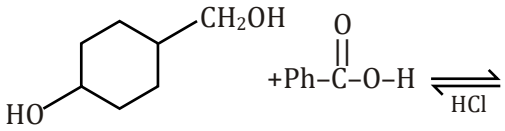
(B) 

(C) 

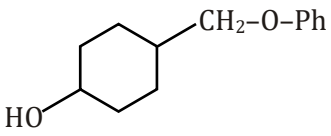
(D) 

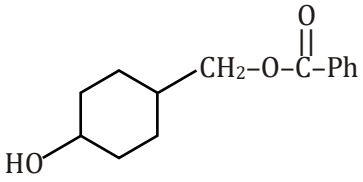
CAE107

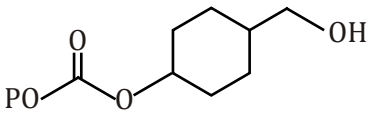
For More Join: @IITJEE_Advanced

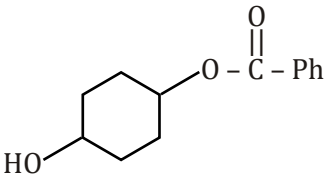
28. 

Major product of above esterification reaction is :

(A) 

(B) 

(C) 

(D) 

CAE108

Oxidation of Alcohol

29. $CH_3CH_2CH_2CH_2OH \xrightarrow{PCC}$ X, X is

(A) $CH_3CH_2CH_2CHO$ (B) $CH_3CH_2CH_2COOH$ (C) CH_3CH_2CHO (D) $CH_3CH_2COCH_3$

CAE109

30. For the conversion $RCH_2OH \longrightarrow RCHO$; most suitable reagent is :

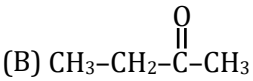
(A) $KMnO_4$ (B) $K_2Cr_2O_7$

(C) CrO_3 (D) Pyridine chlorochromate (PCC)

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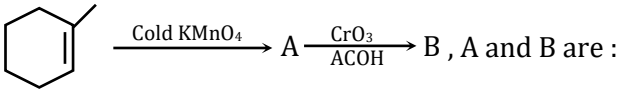
CAE111

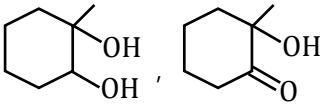
31. An organic compound B is formed by the reaction of ethyl magnesium iodide (CH_3CH_2MgI) with a substance A, followed by treatment with dilute aqueous acid. Compound B does not react with PCC in dichloromethane. Identify A?

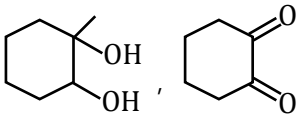
(A)  (B)  (C) $H_2C{=}O$ (D) 

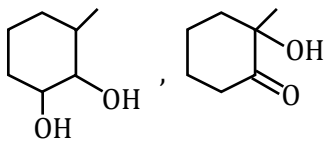
CAE112

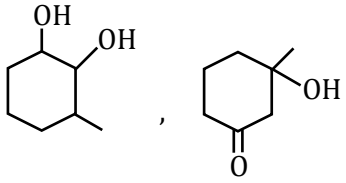
32. For the given reaction sequence



(A) 

(B) 

(C) 

(D) 

CAE113

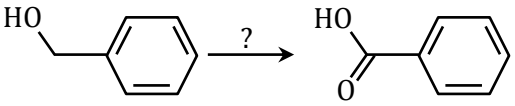
For More Join: @IITJEE_Advanced

33. Which of the following does not react with aqueous Periodic acid ?

(A) $CH_2(OH)-CH_2(OH)$ (B) $CH_2(OH)-C(=O)-CH_2(OH)$ (C) $CH_2(OH)-CH_2-CH_2(OH)$ (D) $CH_2(OH)-CH(OH)-CH_2(OH)$

CAE114

34. Which of the following reagents would not carry out the following reaction ?



(A) PCC (B) $KMnO_4$ (C) $H_2Cr_2O_7$ (D) H_2CrO_4

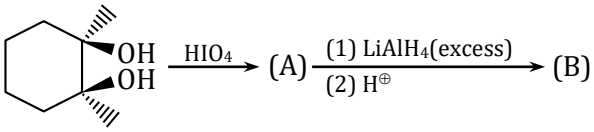
CAE115

35. Which gives 2 moles of HCHO, 1 mole of HCOOH and CO_2 with HIO_4 :

(A) $CH_2(OH)-CH(OH)-C(=O)-CH_2(OH)$ (B) $CH(=O)-CH(OH)-CH(OH)-CH(=O)$ (C) $CH_2(OH)-(CH(OH))_2-CH(=O)$ (D) All of these

CAE116

36.



Total number of stereoisomers of product (B) will be :

(A) 2 (B) 3 (C) 4 (D) 5

CAE117

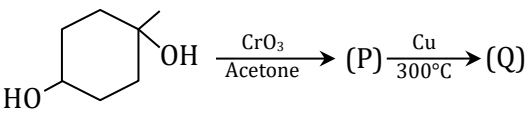
37. $CH(=O)-CH(OH)-CH_2(OH) \xrightarrow{2HIO_4}$ Product (P_1) + Product (P_2)

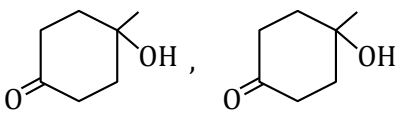
Products obtained in the above reaction are:

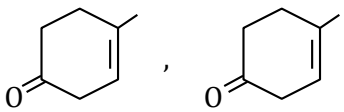
(A) HCHO, HCO_2H (B) HCHO, $2HCO_2H$ (C) CO_2, $2HCO_2H$ (D) CO_2, HCHO, HCO_2H

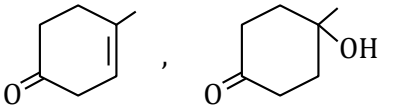
CAE118

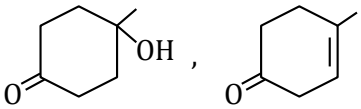
38. (P) and (Q) in the following reaction is:



(A) 

(B) 

(C) 

(D) 

CAE119

Pankers Academy JEE

For More Join: @IITJEE_Advanced

Test of Alcohol

39. Which of the following alcohols gives a red colour in Victor Meyer test?

(A) $CH_3-CH_2-CH_2-OH$

(B) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-OH$

(C) $(CH_3)_3C-OH$

(D) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-CH_3$

CAE121

40. Which of the following does not turn orange colour of chromic acid to green?

(A) 1° alcohol (B) 2° alcohol (C) 3° alcohol (D) Allyl alcohol

CAE122

41. Alcohol which gives blue colouration in Victor Meyer test is :

(A) Only CH_3CH_2—OH

(B) Only CH_3CH_2—CH_2OH

(C) All alcohols containing —CH_2—OH group

(D) All alcohols containing $>$CH–OH group

CAE123

Methode of preparation of ethers and epoxides

42. Williamson's synthesis involves :-

(A) S_N1 mechanism

(B) Nucleophilic addition

(C) S_N2 mechanism

(D) S_E mechanism

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CAE125

43. In the williamson synthesis of ethers given by the general equation $R-X + R'-ONa \longrightarrow R-OR'$, the yield form R–X follows the sequence :-

(A) $CH_3 > 1° > 2° > 3°$

(B) $CH_3 < 1° < 2° < 3°$

(C) $CH_3 < 1° > 2° > 3°$

(D) $CH_3 > 1° < 2° > 3°$

CAE126

44. A on treatment with Na gives B and with PCl_5 gives C. B and C react together to give diethyl ether. A, B and C are in the order :-

(A) C_2H_5OH, C_2H_5Cl, C_2H_5ONa

(B) C_2H_5OH, C_2H_6, C_2H_5Cl

(C) C_2H_5OH, C_2H_5ONa, C_2H_5Cl

(D) C_2H_5Cl, C_2H_6, C_2H_5OH

CAE127

45. A student tried two reactions for preparing tert-butyl ethyl ether:

(I) $C_2H_5ONa + CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-Cl \longrightarrow$

(II) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-ONa + CH_3-CH_2-Cl \longrightarrow$

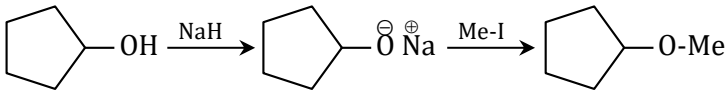
Which reaction will give test-butyl ether?

(A) Only I (B) Only II (C) Both I & II (D) Neither I nor II

CAE128

For More Join: @IITJEE_Advanced

46. The reaction



can be classified as :-

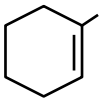
(A) Dehydration reaction
(B) Williamson ether synthesis reaction
(C) Williamson alcohol synthesis reaction
(D) Alcohol formation reaction

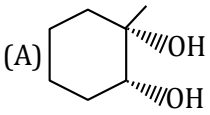
CAE129

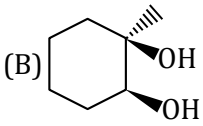
47. The mechanism of formation of ether from dehydration of alcohol doesn't involve :-

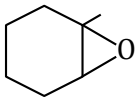
(A) Formation of protonated alcohol.
(B) Nucleophilic attack by alcohol.
(C) Formation of carbocation.
(D) Nucleophilic attack of water molecular.

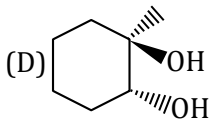
CAE139

48.  $\xrightarrow{mCPBA}$ P, the product P is :

(A) 

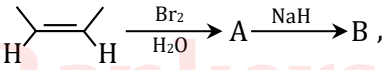
(B) 

(C) 

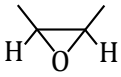
(D) 

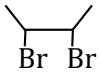
CAE141

Reaction of Alcohol and ether

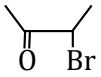
49. 

The product B formed is:

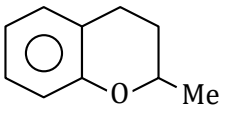
(A) 

(B) 

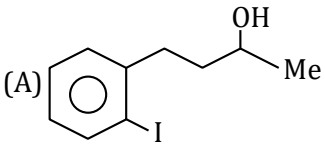
(C) 

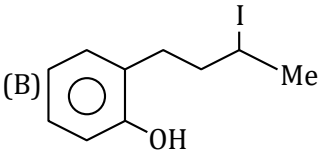
(D) 

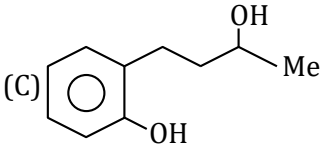
CAE143

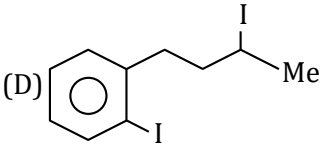
50. 

$\xrightarrow{conc.HI}$ Product ; Product is :

(A) 

(B) 

(C) 

(D) 

CAE144

51. Anisole $\xrightarrow[reflux]{HI(Conc.)}$ Product :-

(A) Ph–I + Me–I
(B) Ph–I + MeOH
(C) Ph–OH + MeI
(D) Ph–OH + MeOH

CAE145

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For More Join: @IITJEE_Advanced

52. $CH_3CH_2-\underset{\underset{CH_3}{|}}{CH}-CH_2-O-CH_2CH_3 + HI \longrightarrow P,$

(A) $CH_3CH_2-\underset{\underset{CH_3}{|}}{CH}-CH_2OH + CH_3CH_2I$

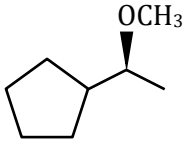
(B) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-CH_2I + CH_3CH_2OH$

(C) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-CH_2I + CH_3-I$

(D) CH_3-CH_2-OH only

CAE146

53. How many total possible alcohols are formed by acidic hydrolysis of following ether.

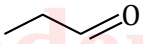


(A) 5 (B) 6 (C) 7 (D) 8

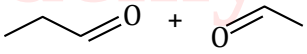
CAE148

54. $CH_3-CH_2-O-\underset{\underset{CH_3}{|}}{C}=CH_2 \xrightarrow{H_3O^{\oplus}}$ Product are

(A)  + EtOH

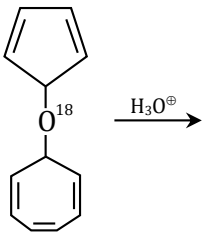
(B)  + EtOH

(C)  + EtOH

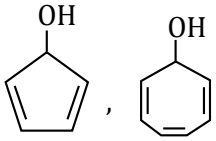
(D) 

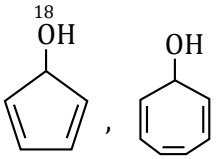
Rankers Academy JEE

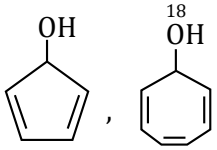
CAE149

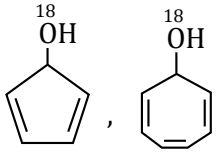
55. 

The products X and Y are

(A) 

(B) 

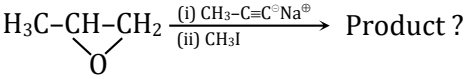
(C) 

(D) 

CAE150

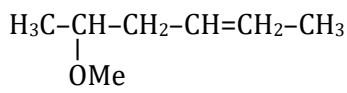
For More Join: @IITJEE_Advanced

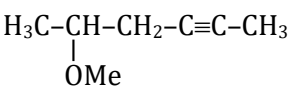
56. For the given reaction

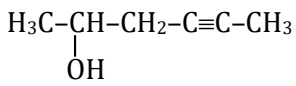


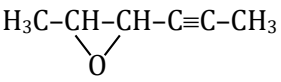
Product ?

Product is :

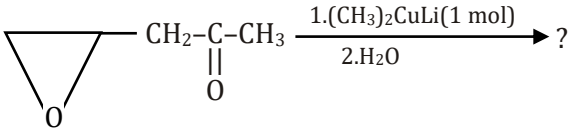
(A) 

(B) 

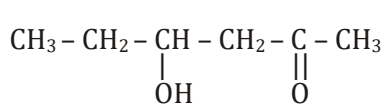
(C) 

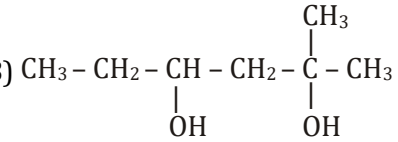
(D) 

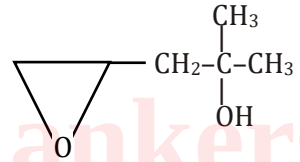
CAE151

57. 

The product is :

(A) 

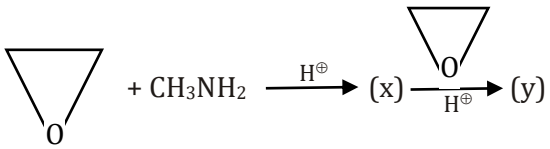
(B) 

(C) 

(D) All

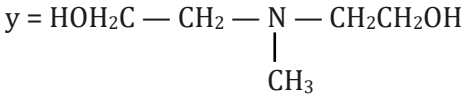
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CAE153

58. 

The products (x) and (y) are :

(A) x = $HOCH_2—CH_2—NH—CH_3$

y = 

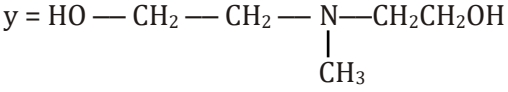
(B) x = $HOCH_2–CH_2–CH_2–NH_2$

y = $HOH_2C–CH_2–CH_2–NH–CH_2–CH_2OH$

(C) x = $HOCH_2–CH_2–NH–CH_3$

y = $HOCH_2–CH_2–CH_2–NH–CH_2–CH_2OH$

(D) x = $HO—CH_2—CH_2—CH_2NH_2$

y = 

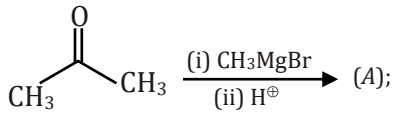
CAE154

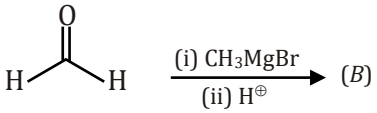
For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Preparation of Alcohols and Ethers

1. 

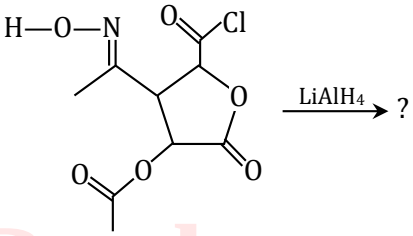


Which of the following reagents can be used to differentiate between A and B?

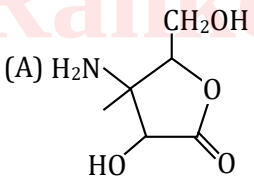
(A) Na (B) $FeCl_3$ (C) Lucas reagents (D) $K_2Cr_2O_7$

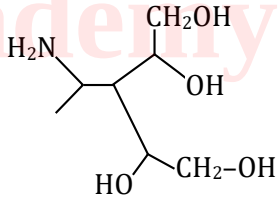
CAE124

2. Find out the possible products in the following reaction:

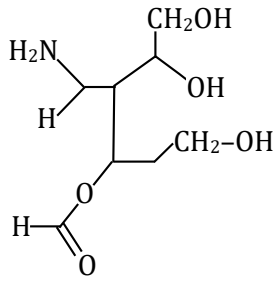


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(A) 

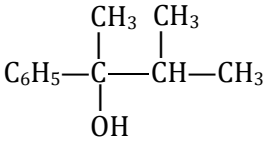
(B) 

(C) CH_3CH_2OH

(D) 

CAE196

3. Which of the following reagents (A to D) would you not select to convert $C_6H_5COCH_3$ (acetophenone) to the following alcohol ?



(A) $CH_3CH_2CH_2MgBr$ and hydrolysis (B) CH_3MgBr and acid hydrolysis

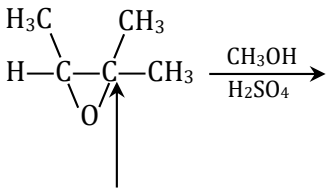
(C) $(CH_3)_2CHMgBr$ and acid hydrolysis (D) PhMgBr and acid hydrolysis

CAE197

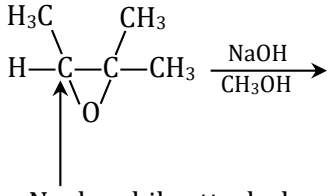
For More Join: @IITJEE_Advanced

Reaction of Alcohols and Ethers

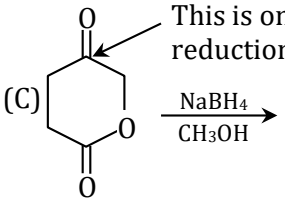
4. Which is/are correct statement?

(A) 

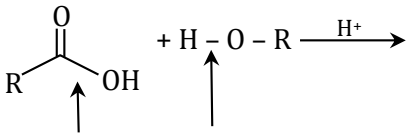
Nucleophile attacks here when epoxy linkage is cleaved

(B) 

Nucleophile attacks here when epoxy linkage is cleaved

(C) 

This is only affected in reduction to 2° alcohol :

(D) 

these bonds are affected in esterification

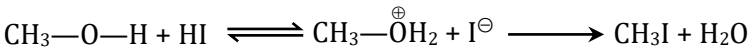
CAE198

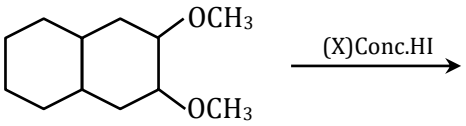
COMPREHENSION TYPE QUESTIONS

Reaction of Alcohols and Ethers

Paragraph for Question No. 5

Here is the mechanism of this reaction, Methyl alcohol reacts to form alkyl halides under acidic conditions by an S_N2 mechanism. In this reaction the function of HI acid is to produce a protonated alcohol. The halide ions then displace a molecule of water from carbon, this produces a methyl iodide.



5. 

x = moles of HI consumed value of x is

(A) 2 (B) 4 (C) 5 (D) 6

CAE199

For More Join: @IITJEE_Advanced

MATRIX MATCH TYPE QUESTIONS

Preparation of Alcohols and Ethers

6. $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH=CH_2 \xrightarrow{\text{Reagent}}$ Alcohol

which is true about alcohol and Reagent?

List - (I) (Alcohol)		List - (II) (Reagent)	
(1)	$CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_2OH$	(P)	B_2H_6, $H_2O_2/NaOH$
(2)	$CH_3-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{OH}{|}}{CH}-CH_3$	(Q)	$Hg(OAc)_2$, H_2O / $NaBH_4$ followed by P.C.C.
(3)	$CH_3-\overset{\overset{OH}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2CH_3$	(R)	$Hg(OAc)_2$, $H_2O/NaBH_4$
(4)	$CH_3-\underset{\underset{CH_3}{|}}{CH}-\overset{\overset{O}{||}}{C}-CH_3$	(S)	dil. H_2SO_4

Pankers Academy JEE

(A) 1 – P, 2-R, 3-S, 4-Q
(B) 1 – P, 2-R, 3-Q, 4-S
(C) 1 – R, 2-S, 3-P, 4-Q
(D) 1 – Q, 2-S, 3-P, 4-R

CAE200

For More Join: @IITJEE_Advanced

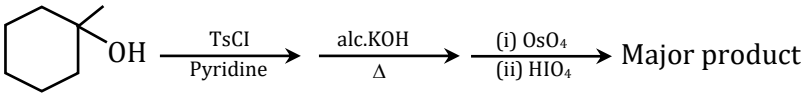
EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Reaction of Alcohols and Ethers

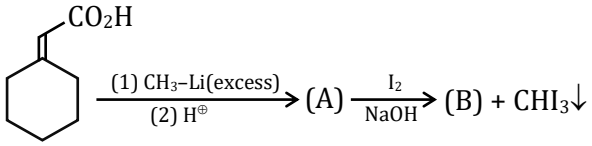
1. Identify the major product of following sequence of reactions:



How many π bonds in major product ?

CAE201

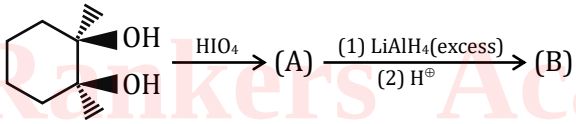
2.



Find D.O.U. in compound (B) ?

CAE202

3.

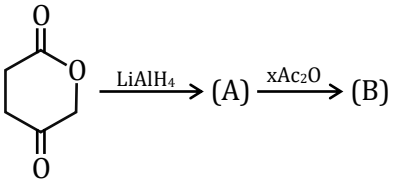


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Total number of stereoisomers of product (B) will be:

CAE156

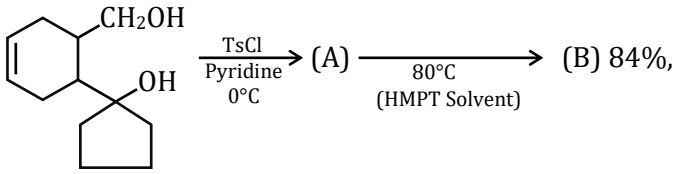
4.



x = moles of anhydride consumed, Find x ?

CAE157

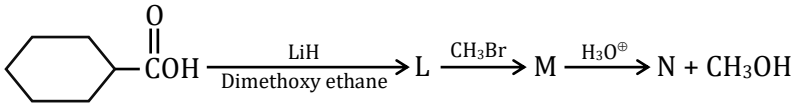
5.



Number of σ bonds in final product (B) ?

CAE203

6.

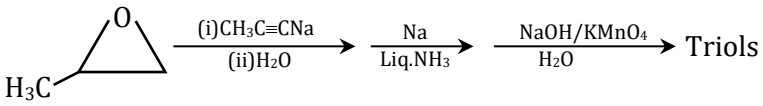


Number of carbon-Oxygen σ-bonds present in product (N) is ?

CAE204

For More Join: @IITJEE_Advanced

7. Consider the reaction given below,



How many different triols isomers are formed?

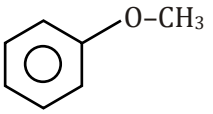
CAE026

8. If propene undergoes oxidation to methyl oxirane. What is the net change in oxidation number of carbon atoms?

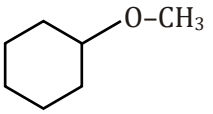
CAE027

9. If (dℓ) or (±) 2-methyl butanoic acid were esterified by reaction with (dℓ) or (±) 2-butanol, how many optically active compounds would be present in the final equilibrium reaction mixture?

CAE028

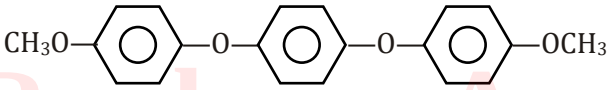
10. 

$\xrightarrow{\text{Conc.HI}}$ (a) mole HI is consumed



$\xrightarrow{\text{Conc.HI}}$ (b) mole HI is consumed Sum of a + b =

CAE029

11. 

How many moles of HI consumed in above reaction ?

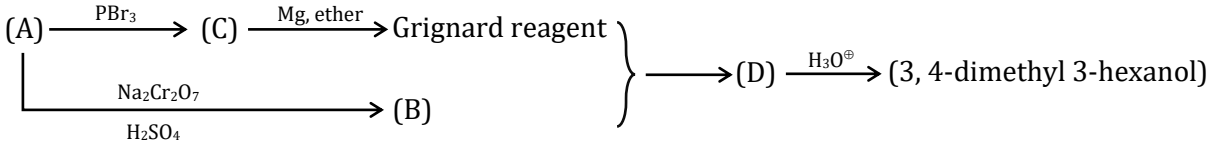
CAE030

Rankers Academy JEE

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Reaction of Alcohols and Ethers

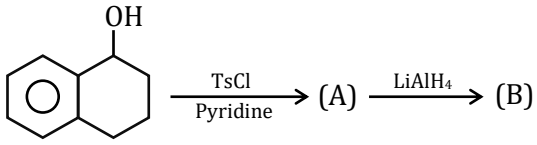
12. 

When Grignard reagent reacts with (B) product (D) will obtained.

If number of π-bonds in compound (B) is x then find $\frac{x}{5}$?

CAE205

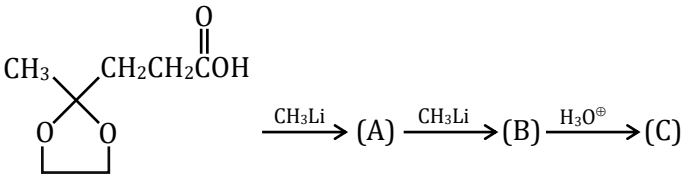
For More Join: @IITJEE_Advanced

13. 

Product (B) of the above reaction is :

If number of oxygen atom in product (B) is x then find (x + 0.59) ?

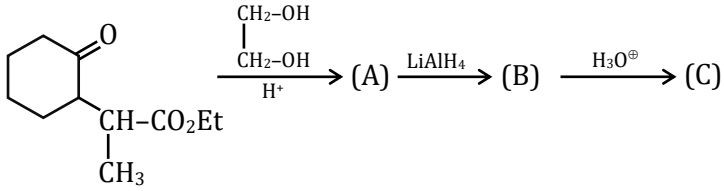
CAE206

14. 

Product (C) of the above reaction is

If number of C–O σ-bonds in product (C) is x then find $\frac{x}{100}$?

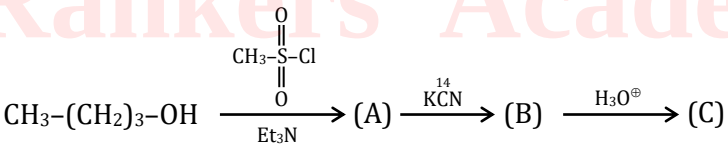
CAE207

15. 

If number of C–O σ-bonds in product (C) is x then find (x^2 – 0.18) ?

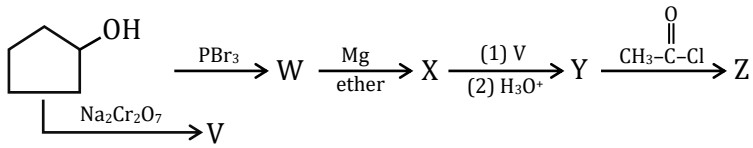
CAE208

Rankers Academy JEE

16. 

If number of O atoms present in product (C) is x then find (x^3 – 0.5) ?

CAE209

17. 

Product Z of above reaction is

If number of D.O.U. in compound (Z) is x then find $\frac{x}{10}$?

CAE210

For More Join: @IITJEE_Advanced

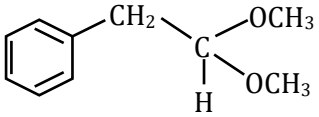
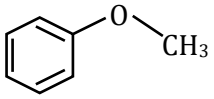
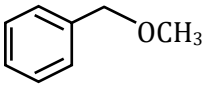
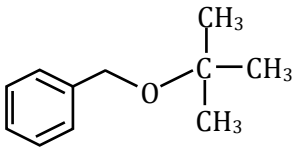
EXERCISE - JEE (Main) PYQ

1. Given below are two statements:
Statement (I): The boiling points of alcohols and phenols increase with increase in the number of C-atoms.
Statement (II): The boiling points of alcohols and phenols are higher in comparison to other class of compounds such as ethers, haloalkanes.
In the light of the above statements, choose the **correct** answer from the options given below:
[JEE (Main) 2025]

(1) Both **Statement I** and **Statement II** are false
(2) **Statement I** is false but **Statement II** is true
(3) **Statement I** is true but **Statement II** is false
(4) Both **Statement I** and **Statement II** are true

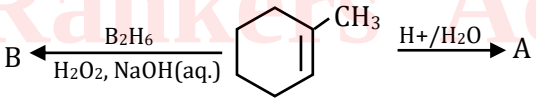
CAE218

2. Which one of the following, with HBr will give a phenol? **[JEE (Main) 2025]**

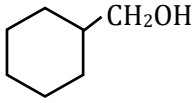
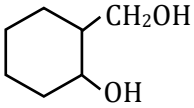
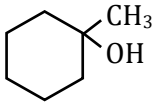
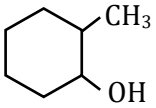
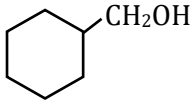
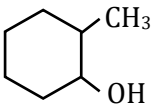
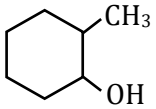
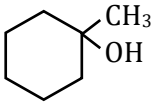
(1) 
(2) 
(3) 
(4) 

CAE219

3. Product A and B formed in the following set of reactions are:



[JEE (Main) 2024]

(1) A =  B = 
(2) A =  B = 
(3) A =  B = 
(4) A =  B = 

CAE211

4. Which among the following has highest boiling point? **[JEE (Main) 2024]**

(1) $CH_3CH_2CH_2CH_3$
(2) $CH_3CH_2CH_2CH_2–OH$
(3) $CH_3CH_2CH_2CHO$
(4) $H_5C_2 – O – C_2H_5$

CAE212

5. For the compounds:

(A) $H_3C–CH_2–O–CH_2–CH_2–CH_3$
(B) $H_3C–CH_2–CH_2–CH_2–CH_3$
(C) $CH_3–CH_2–\underset{\underset{O}{\|}}{C}–CH_2–CH_3$
(D) $H_3C–\underset{\underset{OH}{|}}{CH}–CH_2–CH_2–CH_3$

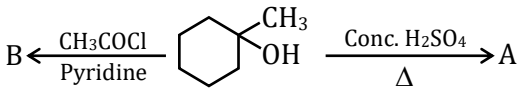
The increasing order of boiling point is :
Choose the **correct** answer from the options given below : **[JEE (Main) 2024]**

(1) (A) < (B) < (C) < (D)
(2) (B) < (A) < (C) < (D)
(3) (D) < (C) < (A) < (B)
(4) (B) < (A) < (D) < (C)

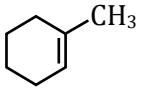
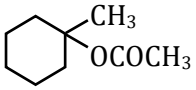
CAE213

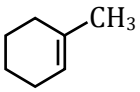
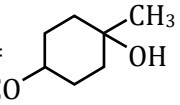
For More Join: @IITJEE_Advanced

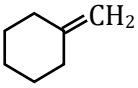
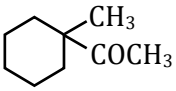
6. Identify the major products A and B respectively in the following set of reactions.

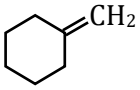
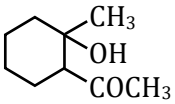


[JEE (Main) 2024]

(1) A=  and B = 

(2) A=  and B = 

(3) A=  and B = 

(4) A=  and B = 

CAE214

7. Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**:

Assertion A: Alcohols react both as nucleophiles and electrophiles.

Reason R: Alcohols react with active metals such as sodium, potassium and aluminium to yield corresponding alkoxides and liberate hydrogen.

In the light of the above statements, choose the ***correct answer*** from the options given below:

[JEE (Main) 2024]

(1) A is false but R is true.

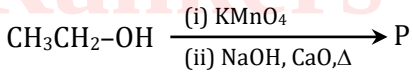
(2) A is true but R is false.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A

CAE215

Rankers Academy JEE

8. 

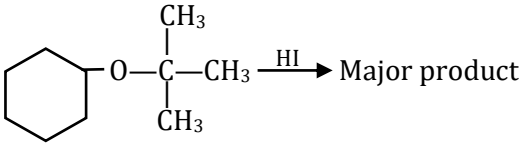
Consider the above reaction sequence and identify the major product P. **[JEE (Main) 2024]**

(1) Methane (2) Methanal

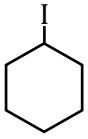
(3) Methoxymethane (4) Methanoic acid

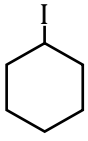
CAE216

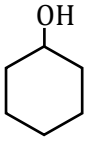
9. Major product formed in the following reaction is a mixture of:

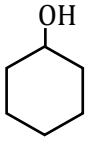


[JEE (Main) 2024]

(1)  and $(CH_3)_3$ CI

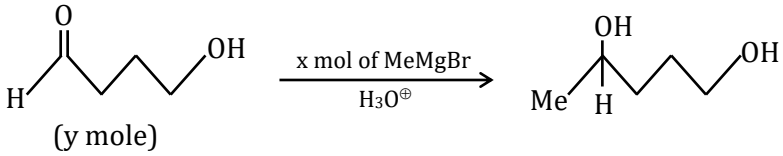
(2)  and $(CH_3)_3$ COH

(3)  and $(CH_3)_3$ COH

(4)  and (CH_3)—C(CH3)—I

CAE217

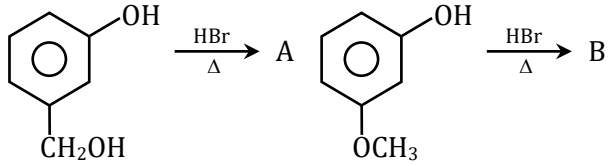
For More Join: @IITJEE_Advanced

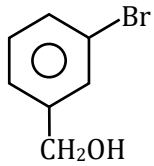
10. 

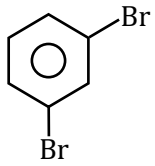
The ratio x/y on completion of the above reaction is _____. **[JEE (Main) 2023]**

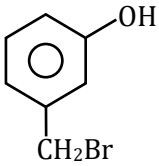
CAE050

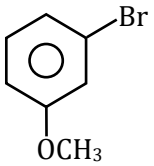
11. 'A' and 'B' formed in the following set of reactions are : **[JEE (Main) 2023]**

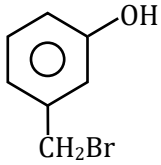


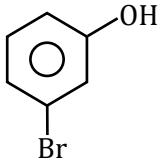
(1) A = 

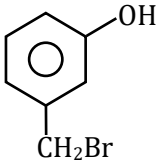
, B = 

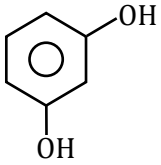
(2) A = 

, B = 

(3) A = 

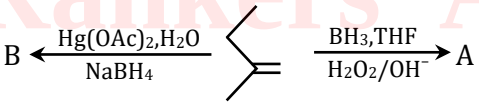
, B = 

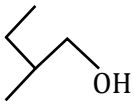
(4) A = 

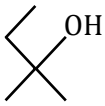
, B = 

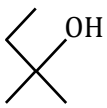
CAE168

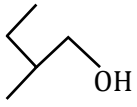
12. Find out the major products from the following reactions. **[JEE (Main) 2023]**

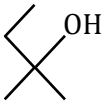


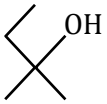
(1) A = 

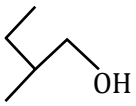
, B = 

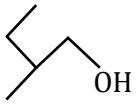
(2) A = 

, B = 

(3) A = 

, B = 

(4) A = 

, B = 

CAE169

13. Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**. **[JEE (Main) 2023]**

Assertion A: Butan–1–ol has higher boiling point than ethoxyethane.

Reason R: Extensive hydrogen bonding leads to stronger association of molecules.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A

(2) A is true but R is false

(3) Both A and R are true but R is not the correct explanation of A

(4) A is false but R is true

CAE172

For More Join: @IITJEE_Advanced

14. Suitable reaction condition for preparation of Methyl phenyl ether is **[JEE (Main) 2023]**

(1) $Ph-Br, MeO^-Na^+$ (2) $PhO^-Na^+, MeOH$

(3) $PhO^-Na^+, MeBr$ (4) Benzene, MeBr

CAE173

15. The product formed in the following multistep reaction is: **[JEE (Main) 2023]**

$$CH_3-CH=CH_2 \xrightarrow[\text{iii) PCC}]{\text{i) } B_2H_6 \text{ ii) } H_2O_2, NaOH} \xrightarrow[\text{iv) } CH_3MgBr]{}$$

(1) $CH_3-CH_2-\underset{OH}{\underset{|}{CH}}-CH_3$ (2) $CH_3-CH_2-CH_2-CH_2-OH$

(3) $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-OCH_3$ (4) $CH_3-\underset{CH_3}{\underset{|}{\overset{OH}{\overset{|}{C}}}}-CH_3$

CAE175

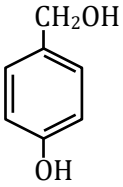
16. Choose the correct option for the following reactions. **[JEE (Main) 2022]**

$$B \xleftarrow[H_2O_2/OH^{\ominus}]{(BH_3)_2} H_3C-\underset{CH_3}{\underset{|}{\overset{CH_3}{\overset{|}{C}}}}-CH=CH_2 \xrightarrow[NaBH_4]{Hg(OAc)_2, H_2O} A$$

(1) 'A' and 'B' are both Markovnikov addition products.
(2) 'A' is Markovnikov product and 'B' is anti-Markovnikov product.
(3) 'A' and 'B' are both anti-Markovnikov products.
(4) 'B' is Markovnikov and 'A' is anti-Markovnikov product.

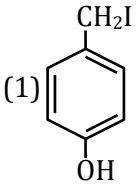
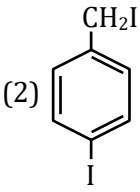
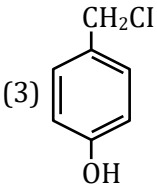
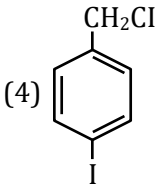
Rankers' Academy JEE

CAE046

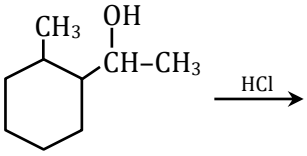
17. 

$$\xrightarrow{\Delta} A \xrightarrow{NaI} B$$

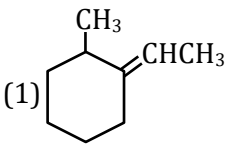
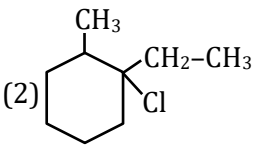
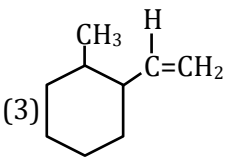
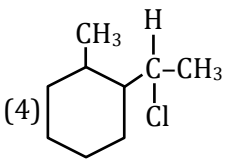
In the above reaction product B is: **[JEE (Main) 2022]**

(1)  (2)  (3)  (4) 

CAE047

18. 

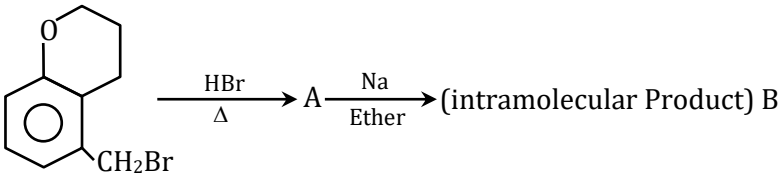
[JEE (Main) 2021]

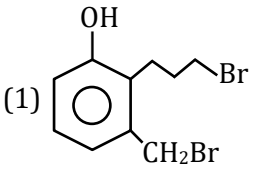
(1)  (2)  (3)  (4) 

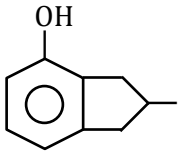
CAE045

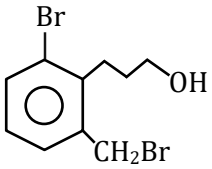
For More Join: @IITJEE_Advanced

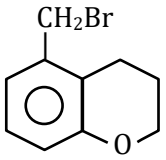
19. In the following reaction sequence, structures of A and B, respectively will be : **[JEE (Main) 2020]**

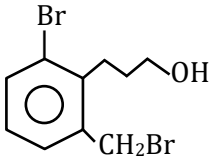


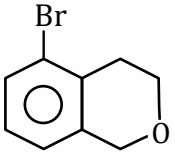
(1) 

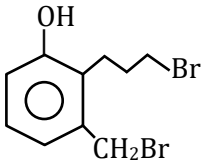
&


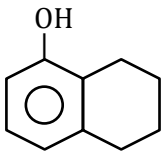
(2) 

&


(3) 

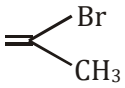
&


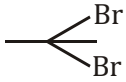
(4) 

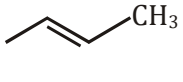
&


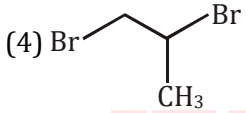
CAE043

20. 1-methyl ethylene oxide when treated with an excess of HBr produces : **[JEE (Main) 2020]**

(1) 

(2) 

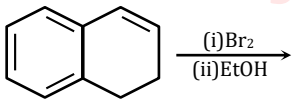
(3) 

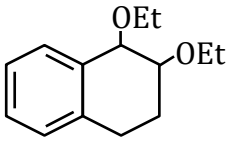
(4) 

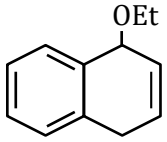
CAE044

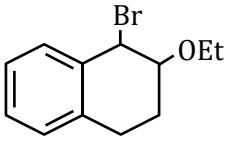
Rankers Academy JEE

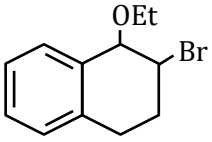
21. The major product the following reaction is : **[JEE (Main) 2019]**



(1) 

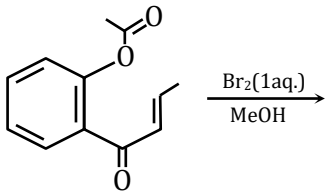
(2) 

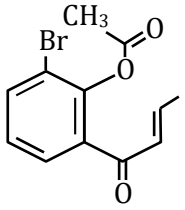
(3) 

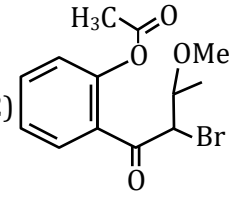
(4) 

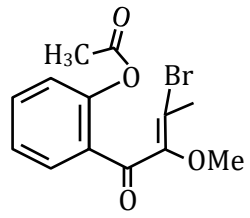
CAE038

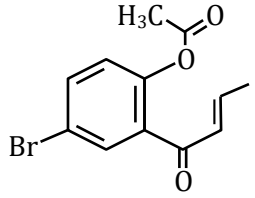
22. The major product obtained in the following conversion is :- **[JEE (Main) 2019]**



(1) 

(2) 

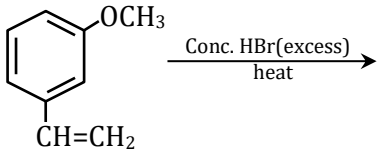
(3) 

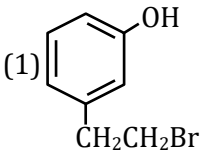
(4) 

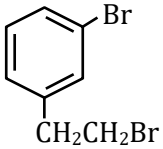
CAE039

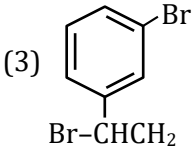
For More Join: @IITJEE_Advanced

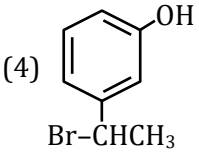
23. The major product of the following reactions: **[JEE (Main) 2019]**



(1) 

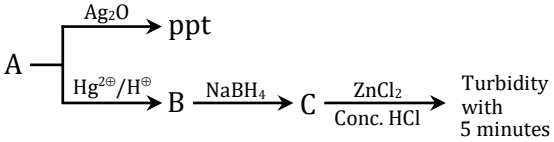
(2) 

(3) 

(4) 

CAE040

24. Consider the following reactions : **[JEE (Main) 2019]**



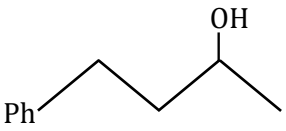
'A' is :

(1) CH≡CH (2) CH_3–C≡CH (3) CH_2=CH_2 (4) CH_3–C≡C–CH_3

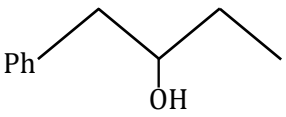
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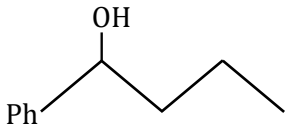
CAE041

25. Heating of 2-chloro-1-phenylbutane with EtOK/EtOH gives X as the major product. Reaction of X with $Hg(OAc)_2/H_2O$ followed by $NaBH_4$ gives Y as the major product. Y is : **[JEE (Main) 2019]**

(1) 

(2) 

(3) 

(4) 

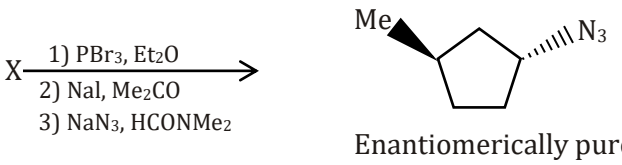
CAE042

For More Join: @IITJEE_Advanced

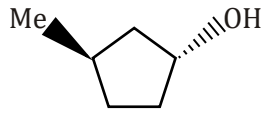
EXERCISE - JEE (Advanced) PYQ

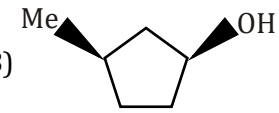
1. In the following reaction sequence, the correct structure(s) of X is (are)

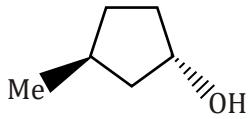
[JEE (Advanced) 2018]

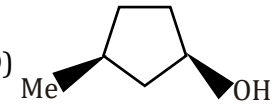


Enantiomerically pure

(A) 

(B) 

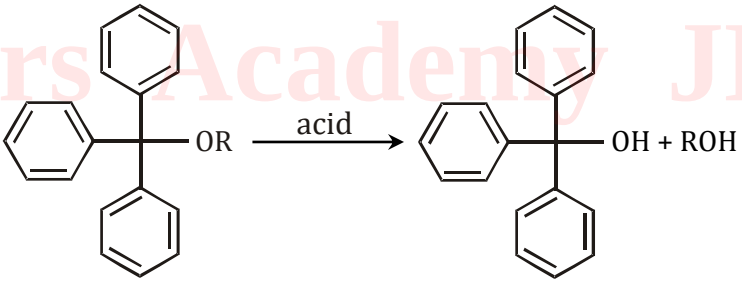
(C) 

(D) 

CAE078

2. The acidic hydrolysis of ether (X) shown below is fastest when : **[JEE (Advanced) 2014]**

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[X]

(A) one phenyl group is replaced by a methyl group

(B) one phenyl group is replaced by a para-methoxyphenyl group

(C) two phenyl groups are replaced by two para-methoxyphenyl group

(D) no structural change is made to X

CAE076

3. The correct combination of names for isomeric alcohols with molecular formula $C_4H_{10}O$ is/are :

[JEE (Advanced) 2014]

(A) tert-butylalcohol and 2-methylpropan-2-ol

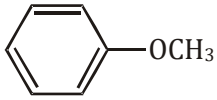
(B) tert- butylalcohol and 1, 1-dimethylethan-1-ol

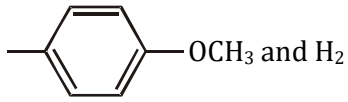
(C) n-butanol and butan-1-ol

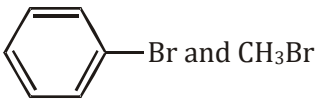
(D) Isobutyl alcohol and 2-methylpropan-1-ol

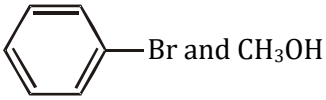
CAE077

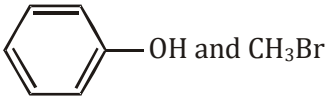
For More Join: @IITJEE_Advanced

4. In the reaction  the products are **[JEE (Advanced) 2010]**

(A) 

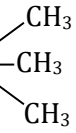
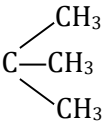
(B) 

(C) 

(D) 

CAE075

5. Phenyl magnesium bromide reacting with t-Butyl alcohol gives **[JEE (Advanced) 2005]**

(A) Ph—OH (B) Ph—H (C)  (D) 

CAE074

6. On acid catalysed hydration, 2-phenyl propene gives: **[JEE (Advanced) 2004]**

(A) 3-phenyl-2-propanol (B) 2-phenyl-1-propanol

(C) 1-phenyl-3-propanol (D) 2-phenyl-2-propanol

CAE073

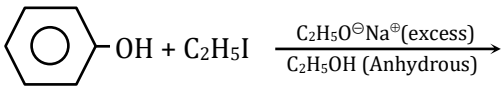
7. Reaction of enantiomerically pure acid with 1 chiral carbon and racemic alcohol with 1 chiral carbon gives an ester which is: **[JEE (Advanced) 2003]**

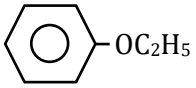
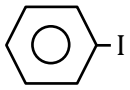
(A) Meso (B) Optically active mixture

(C) Racemic mixture (D) Enantionmerically pure

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CAE072

8.  **[JEE (Advanced) 2003]**

(A)  (B)  (C) $C_6H_5OC_6H_5$ (D) $C_2H_5OC_2H_5$

CAE071

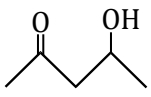
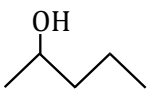
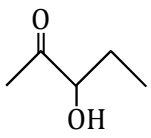
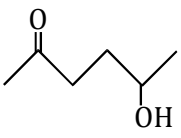
9. Identify the correct order of boiling point of the following compounds: **[JEE (Advanced) 2002]**

$CH_3CH_2CH_2CH_2OH$ (1) $CH_3CH_2CH_2CHO$ (2) $CH_3CH_2CH_2COOH$ (3)

(A) 1 > 2 > 3 (B) 3 > 1 > 2 (C) 1 > 3 > 2 (D) 3 > 2 >1

CAE070

10. Which one of the following will most readily be dehydrated in acidic condition ? **[JEE (Advanced) 2000]**

(A)  (B)  (C)  (D) 

CAE069

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	D	C	A	D	A	D	C	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	B	D	B	C	B	B	B	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	C	C	A	C	A	B	A	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	B	A	C	A	A	B	B	D	A	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	D	C	A	C	B	B	C	C	A	B
Que.	51	52	53	54	55	56	57	58		
Ans.	C	A	D	A	B	B	A	A		

EXERCISE (O-2)

Que.	1	2	3	4	5	6
Ans.	CD	BC	ABD	ABCD	C	A

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EXERCISE (S)

1.	2	**2.**	3	**3.**	3	**4.**	3	**5.**	33
6.	2	**7.**	4	**8.**	2	**9.**	4	**10.**	3
11.	2	**12.**	0.2	**13.**	0.59	**14.**	0.02	**15.**	3.82
16.	7.5	**17.**	0.3						

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	2	2	2	2	1	4	1	4	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	1	1	3	1	2	1	2	4	4
Que.	21	22	23	24	25					
Ans.	4	2	4	2	4					

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	ACD	D	B	D	B	D	B	A

For More Join: @IITJEE_Advanced